

Search for Pair Production of Higgs Bosons via Vector Boson Fusion

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Ruta de la presentación

- 1 Réplica de los yields de Thiziri para $\kappa_{2V} = 0$
- 2 Estudio preliminar del working point

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Objetivo de la réplica

- El objetivo de esta parte fue reproducir los **yields en la región señal (SR)** reportados por Thiziri para:

$$\kappa_{2V} = 0$$

en el modo **DEFAULT**.

- En esta etapa se estudió únicamente la **SR central elíptica**, comparando:
 - **SR baseline**
 - **SR optimizada**
- Las definiciones de **VR** y **CR** asociadas a los métodos *Quadrant* y *Ring* se dejan para un estudio posterior.

Muestras y selección utilizadas

- Se trabajó con la combinación de campañas:

$$\text{MC20a} + \text{MC20d} + \text{MC20e} + \text{MC23a} + \text{MC23d}$$

consistente con la configuración usada por Thiziri (*MC20 + MC23, sin mc23e*).

- Muestra de señal:

$$\text{VBF } HH, \quad \kappa_{2V} = 0$$

- Muestras de fondo:

$$t\bar{t} + \text{QCD multijet}$$

- **Tagger utilizado:** GN3XPV01 con **WP = 0.4**
- Se impuso el requisito de **2Tag / 2Pass**.

Cortes implementados y definición de S y B

Selección explícita implementada

$$p_T(H_1) > 520 \text{ GeV}, \quad p_T(H_2) > 250 \text{ GeV}$$

$$m_{jj} > 400 \text{ GeV}, \quad \Delta\eta_{jj} > 2$$

$$2\text{Tag} / 2\text{Pass}$$

Definición de la región señal

Los eventos deben quedar dentro de la elipse:

$$D = \sqrt{\left(\frac{m_{H1}(m_{H1} - c_1)}{p_1}\right)^2 + \left(\frac{m_{H2}(m_{H2} - c_2)}{p_2}\right)^2} < D_{\text{cut}}$$

Parámetros de las elipses usadas

Baseline SR	
Parámetro	Valor
c_1	124.0
c_2	117.0
p_1	1500
p_2	1900
D_{cut}	1.6

SR optimizada de Thiziri	
Parámetro	Valor
c_1	126.9
c_2	119.7
p_1	1334
p_2	1506
D_{cut}	2.0

- La primera tabla corresponde a la definición **baseline** de la SR.
- La segunda tabla corresponde a la SR **optimizada** reportada por Thiziri para $\kappa_{2V} = 0$ en **DEFAULT**.

- S = yield esperado de señal dentro de la SR.
- B = yield esperado de fondo dentro de la misma SR.
- Se usó el peso completo por evento:

$$w_{\text{full}} = mc_sf \times luminosity_boosted \times pileupweight_boosted$$

Resultado de la réplica para $\kappa_{2V} = 0$

=== Resultados SR, $\kappa_{2V}=0$, DEFAULT, selección explícita de Thiziri, peso completo ===

--- Baseline SR ---

$S = 34.803598$

$B = 10.243647$

$S/(S+B) = 7.726021e-01$

--- Optimised SR ---

$S = 36.132430$

$B = 8.262305$

$S/(S+B) = 8.138900e-01$

=== Referencia de Thiziri (DEFAULT, $\kappa_{2V}=0$, SR) ===

Baseline SR: $S = 34.804$, $B = 10.24$, $S/(S+B) = 0.773$

Optimised SR: $S = 36.136$, $B = 8.26$, $S/(S+B) = 0.814$

- La réplica reproduce con muy buen acuerdo los valores reportados por Thiziri para la SR en **DEFAULT**.
- Esto valida la combinación de muestras, la selección explícita implementada, el uso de GN3XPV01 con $WP = 0.4$ y la normalización con peso completo.

Ruta de la presentación

- 1 Réplica de los yields de Thiziri para $\kappa_{2V} = 0$
- 2 Estudio preliminar del working point

Estudio preliminar del working point

- Se realizó un barrido preliminar de los working points disponibles del tagger GN3XPV01:

0,08, 0,15, 0,25, 0,4, 0,6

- Se mantuvo fija la selección explícita de Thiziri en **DEFAULT** y se compararon:
 - SR baseline
 - SR optimizada
- Para cada WP se calcularon S , B , $S/(S + B)$ y Z_A .

Tablas obtenidas para el escaneo en WP

```

=== Baseline ===
      WP      S      B      S_over_B      S_over_SplusB      S_over_sqrtB      \
0  0p08  12.595015  0.059666  211.090868  0.995285  51.562513
1  0p15  19.587711  0.149560  130.969102  0.992422  50.649629
2  0p25  27.064462  0.331532  8.123730  0.890396  14.827824
3  0p4   34.803598  10.243647  3.397579  0.772602  10.874188
4  0p6   41.593993  20.938085  1.986523  0.665162  9.089963

      Z_A
0  10.506788
1  12.392002
2  8.959620
3  7.989219
4  7.324339

```

```

=== Optimised ===
      WP      S      B      S_over_B      S_over_SplusB      S_over_sqrtB      \
5  0p08  12.991300  0.019928  651.920998  0.998468  92.028808
6  0p15  20.269114  0.099779  203.140252  0.995101  64.167538
7  0p25  28.087425  0.125492  8.986561  0.899865  15.887396
8  0p4   36.132430  0.262305  4.373166  0.813890  12.570327
9  0p6   43.218638  0.947510  2.280967  0.695212  9.928760

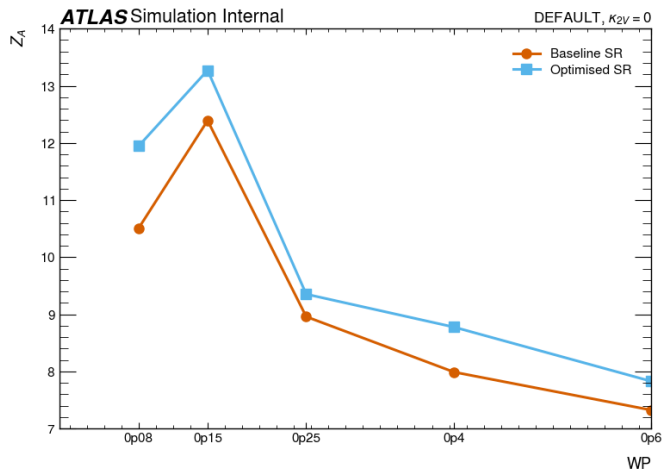
      Z_A
5  11.944907
6  13.271707
7  9.353181
8  8.776519
9  7.828580

```

Figura: Escaneo de WP para la SR baseline.

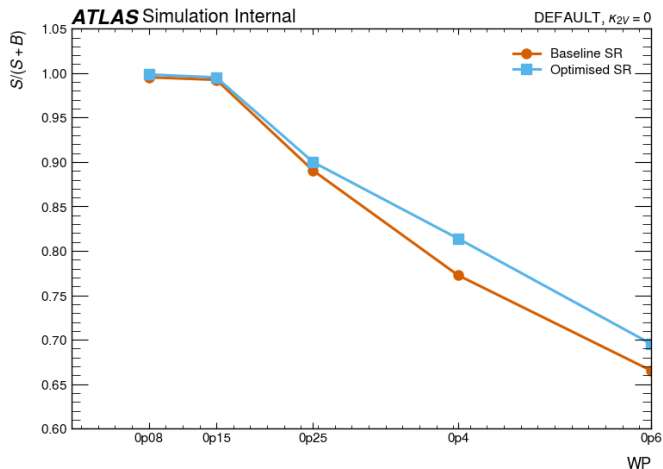
Figura: Escaneo de WP para la SR optimizada.

Comportamiento de Z_A con el WP



- La SR optimizada mejora respecto a la baseline para todos los WPs estudiados.
- En este estudio preliminar, el máximo de Z_A se obtiene para un WP más *tight*.

Pureza $S/(S+B)$ en función del WP



- Los WPs más *tight* reducen fuertemente el fondo y aumentan la pureza.
- La SR optimizada mantiene una pureza mayor que la baseline en todos los WPs estudiados.

Objectives

- **Implementation of the Rings Method:** Extend the central elliptical Signal Region (SR) optimization
- **Comparative Performance across Jet Configurations:** Validate and run the distance discriminant algorithm using
 - **DEFAULT** mode (kinematic raw jet masses without corrections).
 - **BJR10v01** mode (incorporating muon-in-jet and b-jet regression energy corrections).
- **Tagger Working Point and Optimization:** Perform a systematic scan over the available GN3XPV01 tagger working points (0,08, 0,15, 0,25, 0,4, 0,6) to maximize the Asimov statistical significance (Z_A).
- **Validation of External Regions for Background Modeling:** Quantify the signal contamination in the Validation Region (VR) and Control Region (CR) to certify the feasibility of a future data-driven background estimation framework.

Region Definitions: The Rings Method

- To model the multijet QCD background, a custom log-adaptive metric (D_{custom}) is utilized to construct concentric outer regions around the Higgs mass centers (c_1, c_2):

$$D_{\text{custom}} = \sqrt{\left(\frac{m_{H1} - c_1}{0,1 \times \ln(m_{H1})}\right)^2 + \left(\frac{m_{H2} - c_2}{0,1 \times \ln(m_{H2})}\right)^2}$$

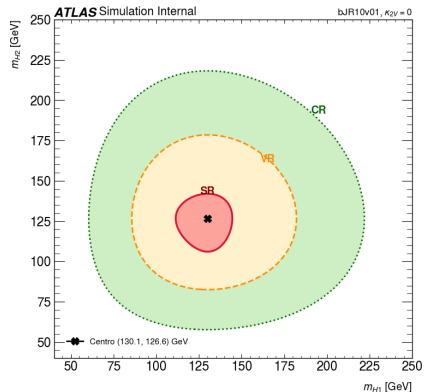
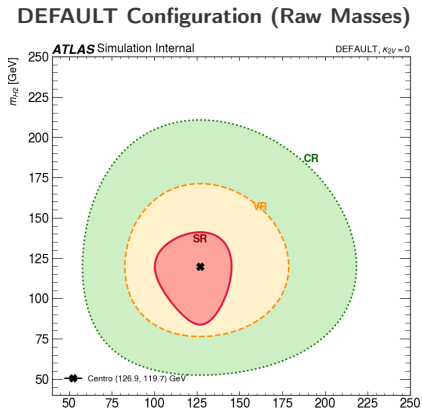
Geometric Boundaries

- **Validation Region (VR):** Outside SR, bounded by $D_{\text{custom}} < 100,0$.
- **Control Region (CR):** Bounded by $100,0 \leq D_{\text{custom}} < 170,0$.

Methodology Remarks

- Metrics scale adaptively to track the QCD background kinematic tail.

bjR10v01 Configuration (With Regression)



DEFAULT Configuration: Signal Region (SR) Results

SR Yields and Asimov Significance (Z_A)

=== PROCESAMIENTO DEFAULT COMPLETADO ===

	WP	Region	S	B	Z_A
0	0p08	SR	12.991300	0.019928	11.944907
3	0p15	SR	20.269114	0.099779	13.271707
6	0p25	SR	28.087425	3.125492	9.353181
9	0p4	SR	36.132430	8.262305	8.776519
12	0p6	SR	43.218638	18.947510	7.828580

Region	S	B	$S/(S+B)$	Z_A
SR (Opt)	36.136	8.26	0.814	8.777

SR Observations

- **Optimal Sensitivity:** The maximum statistical significance is achieved at a tighter working point, specifically **WP = 0p15** ($Z_A = 13, 27$).

DEFAULT Configuration: VR & CR Multi-Region Validation

VR & CR Multi-Region Yields Summary

=== TABLA RESUMEN: REGION DE VALIDACION (VR) - MODO DEFAULT ===

WP	S	B	Contaminacion %
0p08	3.517017	0.567158	19.736226
0p15	5.460000	2.915269	19.737373
0p25	7.583998	10.145283	19.838081
0p4	9.763163	25.047896	19.880020
0p6	11.754354	57.730180	19.971343

=====

=== TABLA RESUMEN: REGION DE CONTROL (CR) - MODO DEFAULT ===

WP	S	B	Contaminacion %
0p08	1.311791	2.632521	7.361295
0p15	1.934143	10.081939	6.991739
0p25	2.558070	26.379129	6.691351
0p4	3.214838	60.856994	6.546141
0p6	3.883109	140.104801	6.597632

Methodological Implications

- The signal leakage fraction remains perfectly stable across all WPs ($\sim 19.8\%$ in VR, $\sim 6.6\%$ in CR), validating the spatial consistency of the Rings geometry.

VR (Opt)	9.763	25.05	0.280	1.841
VR (BL)	11.748	22.18	0.346	2.312
CR (Opt)	3.212	60.86	0.050	0.408

bJR10v01 Configuration: Signal Region (SR) Results

SR Yields and Asimov Significance (Z_A)

	WP	Region	S	B	Z_A
0	0p08	SR	13.664956	0.105736	10.333377
3	0p15	SR	20.860261	0.110522	13.352568
6	0p25	SR	28.511071	1.016412	11.913712
9	0p4	SR	36.121195	2.716553	11.592096
12	0p6	SR	42.755726	12.285478	8.920493

Region	S	B	$S/(S+B)$	Z_A
SR (Opt)	35.638	1.76	0.953	12.542

Regression Impact on SR

- **Highest Global Sensitivity:** Incorporating b-jet regression energy corrections shifts the maximum significance to a massive $Z_A = 13,35$ at **WP = 0p15**.
- **Resolution Peak:** Correcting for semileptonic neutrino losses tightens mass distributions, significantly outperforming the DEFAULT mode sensitivity across all tight working points.

bjR10v01 Configuration: VR & CR Validation

VR & CR Multi-Region Yields Summary

=== TABLA RESUMEN: REGION DE VALIDACION (VR) - MODO bJR10v01 ===

WP	S	B	Contaminacion %
0p08	3.995550	0.837453	22.448807
0p15	6.455576	2.854018	23.393470
0p25	9.120776	11.740070	23.932704
0p4	12.091085	31.465291	24.717696
0p6	14.853161	68.085745	25.339648

=====

=== TABLA RESUMEN: REGION DE CONTROL (CR) - MODO bJR10v01 ===

WP	S	B	Contaminacion %
0p08	0.137989	2.159653	0.775283
0p15	0.279793	10.285514	1.013905
0p25	0.478249	30.030169	1.254915
0p4	0.704436	64.451696	1.440072
0p6	1.007401	141.857766	1.718637

Background Modeling Status

- **Geometric Alignment:** The concentric boundaries successfully accommodate the narrower mass constraints of the b-jet regression calibration.

VR (Opt)	11.664	26.97	0.302	2.108
VR (BL)	11.189	25.51	0.305	2.077
CR (Opt)	0.701	59.56	0.012	0.091

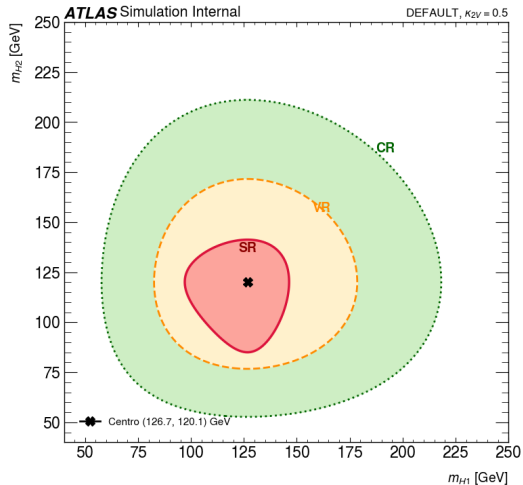
Region Definitions & Mass Planes ($\kappa_{2V} = 0,5$)

- To model the multijet QCD background, a custom log-adaptive metric (D_{custom}) constructs concentric outer regions around the Higgs mass centers (c_1, c_2):

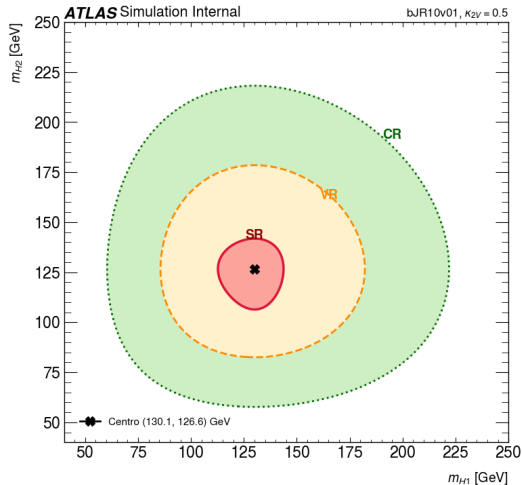
$$D_{\text{custom}} = \sqrt{\left(\frac{m_{H1} - c_1}{0,1 \times \ln(m_{H1})}\right)^2 + \left(\frac{m_{H2} - c_2}{0,1 \times \ln(m_{H2})}\right)^2}$$

- **Annulet Cuts: Validation Region (VR)** bounded by $D_{\text{custom}} < 100,0$, and **Control Region (CR)** bounded by $100,0 \leq D_{\text{custom}} < 170,0$.

DEFAULT Configuration (Raw Masses)



bJR10v01 Configuration (With Regression)



DEFAULT Configuration: Multi-Region Comparison ($\kappa_{2V} = 0,5$)

My Results (SR, VR & CR)

=== PROCESAMIENTO DEFAULT COMPLETADO PARA K2V = 0.5 ===

--- REGION DE SEÑAL (SR) ---

WP	Region	S	B	Z_A
0p08	SR	3.459238	0.059898	4.663771
0p15	SR	5.398937	0.139750	5.473856
0p25	SR	7.453643	3.401816	3.207050
0p4	SR	9.591504	8.645934	2.835732
0p6	SR	11.493008	20.621570	2.337890

--- REGION DE VALIDACION (VR) ---

WP	S	B	Contaminacion %
0p08	0.795377	0.527188	17.340458
0p15	1.249062	2.875298	17.473593
0p25	1.716505	9.868958	17.464133
0p4	2.248316	23.818504	17.700544
0p6	2.758321	55.068713	18.030342

--- REGION DE CONTROL (CR) ---

WP	S	B	Contaminacion %
0p08	0.332211	2.681347	7.242721
0p15	0.500284	10.116721	6.998659
0p25	0.658594	26.268992	6.700695
0p4	0.862139	61.593702	6.787452
0p6	1.046889	140.293058	6.843210

Thiziri's Reference

Region	S	B	S/(S+B)	Z_A
SR (Opt)	9.588	8.47	0.531	2.857
SR (BL)	9.038	10.24	0.469	2.512
VR (Opt)	2.252	24.00	0.086	0.453
VR (BL)	2.994	22.18	0.119	0.622
CR (Opt)	0.8629	61.59	0.014	0.110
CR (BL)	0.6913	66.16	0.010	0.085

DEFAULT

bjR10v01 Configuration: Multi-Region Comparison ($\kappa_{2V} = 0,5$)

My Results (SR, VR & CR)

--- REGION DE SEÑAL (SR) ---

WP	Region	S	B	Z_A
0p08	SR	3.409893	0.105736	4.221127
0p15	SR	5.242638	0.110521	5.572948
0p25	SR	7.143751	0.570562	5.088408
0p4	SR	9.050277	2.270702	4.274962
0p6	SR	10.724390	8.177596	3.197810

--- REGION DE VALIDACION (VR) ---

WP	S	B	Contaminacion_%
0p08	1.129892	0.854085	24.674780
0p15	1.808366	2.970997	25.377138
0p25	2.536579	12.310902	25.921071
0p4	3.408125	32.029613	26.961354
0p6	4.230132	71.539440	27.792956

--- REGION DE CONTROL (CR) ---

WP	S	B	Contaminacion_%
0p08	0.039353	2.143021	0.859404
0p15	0.074961	10.168536	1.051945
0p25	0.105449	29.923711	1.077575
0p4	0.182375	64.367116	1.442751
0p6	0.265635	142.635789	1.745283

Thiziri's Reference

Region	S	B	S/(S+B)	Z_A
SR (Opt)	8.968	1.74	0.837	4.582
SR (BL)	9.442	6.21	0.603	3.170
VR (Opt)	3.271	27.13	0.108	0.616
VR (BL)	2.857	25.51	0.101	0.555
CR (Opt)	0.186	59.61	0.003	0.024
CR (BL)	0.149	65.85	0.002	0.018

bjR10v01

Factor de Escala y Estabilidad

- Factor de escala constante en todas las regiones.
- Explicación de la contaminación plana en VR y CR. ????

```
=== ANÁLISIS DE FACTORES DE ESCALA (SEÑAL DEFAULT) ===  
      Salto_WP  Factor_SR  Factor_VR  Factor_CR  
Base (0p08)    1.0000    1.0000    1.0000  
0p08 -> 0p15   1.5607    1.5704    1.5059  
0p15 -> 0p25   1.3806    1.3742    1.3164  
0p25 -> 0p4    1.2868    1.3098    1.3091  
0p4 -> 0p6     1.1982    1.2268    1.2143
```

Figura: Estabilidad del factor multiplicador.

Histograma de Densidad (D_{custom})

- Normalización a área = 1
- Superposición e invariancia de forma en los 5 Working Points.
- Validación visual de la independencia cinemática.

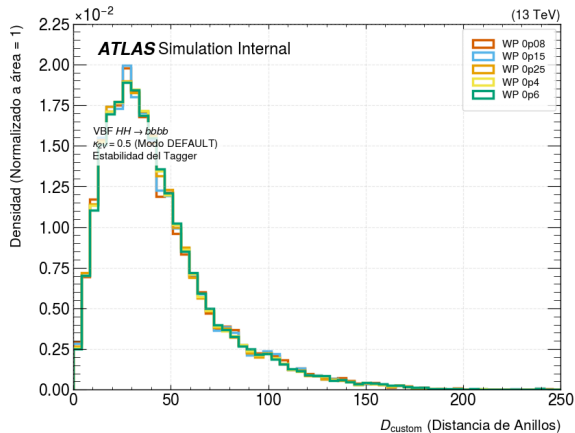


Figura: Distribución de D_{custom} .

Análisis Contaminación (WP 0p4)

